

Uncertainty-Guided Spreading Activation for Efficient Patch Selection in Vision Graphs

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(a) Uniqueness of the Topic:

- This project combines spatial graph diffusion, neuroscience-inspired spreading activation, and information-theoretic uncertainty estimation to select the most predictively critical image patches for classification. Unlike DynamicViT and EViT which use attention scores as heuristic selection signals, our method identifies patches by measuring predictive entropy spikes when patches are removed — a theoretically grounded criterion rooted in mutual information. No existing work combines spatial graph structure, diffusion dynamics, and uncertainty-guided pruning in a single unified vision pipeline.

(b) Learning Goal:

- We aim to understand whether spatially coherent patch selection via graph diffusion can match full-transformer accuracy at significantly reduced FLOPs, while building hands-on expertise in Graph Neural Networks (GCN, GIN, ViG-style aggregation), ViT feature extraction, and Bayesian uncertainty estimation

(c) Possible Datasets:

- CIFAR-10 and CIFAR-100: <https://www.cs.toronto.edu/~kriz/cifar.html>
- Backbone: ViT-Tiny pretrained on ImageNet-1K via timm (<https://github.com/huggingface/pytorch-image-models>)

(d) Reference Paper:

- Rao et al. "DynamicViT: Efficient Vision Transformers with Dynamic Token Sparsification." *NeurIPS*, 2021.
- Han et al. "Vision GNN: An Image is Worth Graph of Nodes." *NeurIPS*, 2022.
- Liang et al. "EViT: Expediting Vision Transformers via Token Reorganizations." *ICLR*, 2022.
- Dosovitskiy et al. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." *ICLR*, 2021.
- Anderson, J. R. (1983). A spreading activation theory of memory. *Journal of Verbal Learning and Verbal Behavior*, 22(3), 261–295 – *spreading paper*